

Linkage

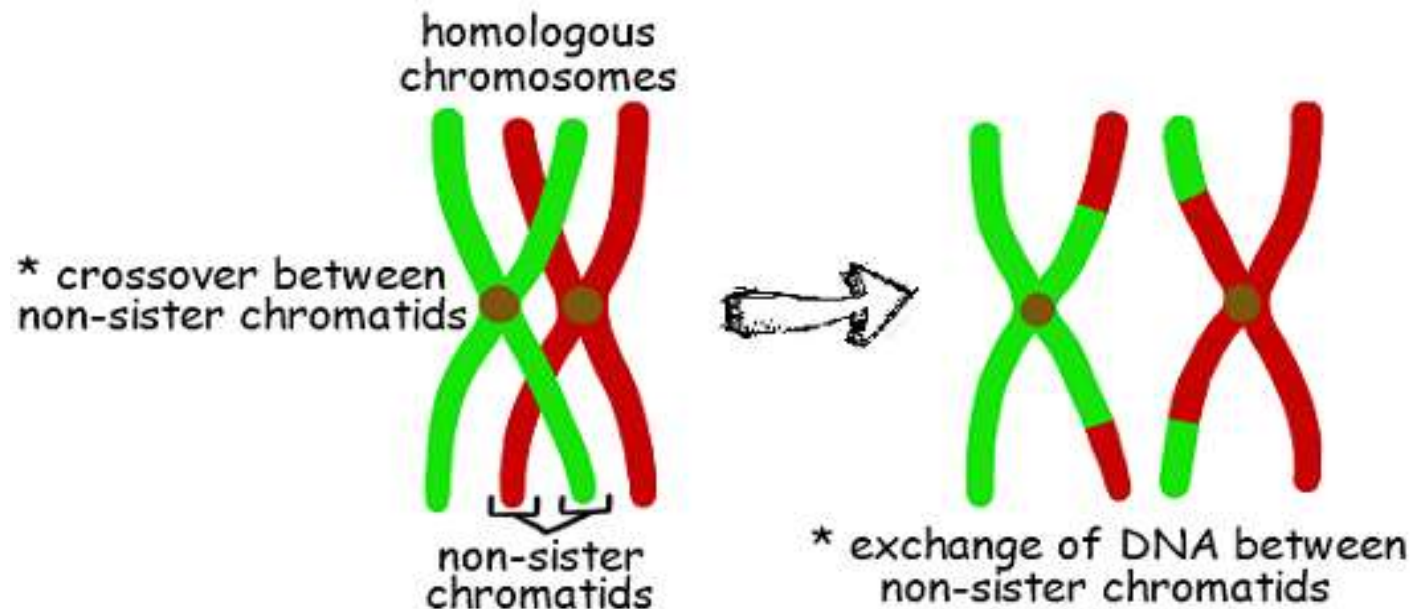
Linked genes:

- Most chromosomes consist of hundreds or thousands of genes.
- Genes located physically close together on the same chromosome are called linked genes. These genes travel together to the same gametes during meiosis. These genes are not transmitted according to the law of independent assortment but they tend to separate by crossing over if they are far enough apart .

Linkage explains why certain characteristics are frequently inherited together e.x. blonde hair with blue eyes.

Crossing over:

- The homologous chromosomes are paired in synapsis during prophase of meiosis I, non sister chromatids may break at distinctive points and switch fragments.
- An exchange of chromosomal arms of paternal and maternal chromatids accounts for the recombination of linked genes.



Genetic map

-The discovery of linked genes and recombination due to crossing over led Morgan to a method for constructing genetic maps. A genetic map lists the sequence of gene's loci along a particular chromosome. Genetic map Shows the relative position of alleles on a chromosome and the distance between genes(measured by the map unit or centi- Morgan)

Genetic maps of human chromosomes are of great importance to our own welfare.

The knowledge of where particular genes are located on human chromosomes can often be used to tell if a fetus carries a genetic disorder for which it is at risk. The hope is that knowledge of what is different at the gene level may suggest a successful therapy for particular genetic disorders.

Physical map:

Is more accurate than the genetic map, so it determines the actual distances between genes on a chromosome by measuring the DNA base pairs.



GENETIC MAPPING

NHGRI FACT SHEETS

genome.gov

Genetic Map



Cytogenetic Map



Physical Map



DNA Sequence

....GATCTGCATGCATGCTAGCTAGCTAGCTAGCTAGAGCTTCG....

Bases



NIH

National Human Genome
Research Institute

-Gene mapping using molecular analysis of DNA:

Human genes may be analyzed in relation to DNA markers establishing their positions along chromosomes.

For example, the location of DNA markers referred to as microsatellites, minisatellites, and restriction fragment length polymorphisms(RFLPs) can be mapped on specific chromosomes. Then a gene such as the one involved in cystic fibrosis, can be located in relation to such markers.

Sex determination:

-Human beings have 22 pairs of **autosomes**, which are structurally the same in both sexes (homologous). One special pair of chromosomes is called the **sex chromosomes** because this pair determines the sex of the individual. The sex chromosomes in the human female are **XX** and those in the male are **XY**. The male determines the sex of the new individual.

Sexual differentiation in humans

During early development , every human embryo undergoes a period when it is potentially hermaphroditic(bipotential gonads). By the 5th week of gestation, gonadal primordia(the tissue that will form the gonad) arise as a pair of ridges associated with each embryonic kidney.

Primordial germ cells migrate to these ridges, Where an outer **cortex** and inner **medulla** formed. The cortex is capable of developing into an ovary, while the inner medulla may develop into a testis.

If the cells of the genital ridge have the XY constitution (in presence of SRY gene), development of the medullary region into a testis is initiated around 7th week. However, in absence of the Y chromosome, no male development occurs, and the cortex of the genital ridge subsequently forms ovarian tissue under the actions of estrogen hormone.

Parallel development of the appropriate male(Wolffian) or female(müllerian) duct systems degenerates.

Once male differentiation is initiated , the embryonic testicular tissue secretes two hormones(Testosterone and anti müllerian hormones that are essential for continued male sexual differentiation.

- Testosterone hormone secreted by Leydig cells which stimulates the development of epididymis ,vas deferens ,seminal vesicle, and ejaculatory duct.

- Anti müllerian hormone secreted by Sertoli cells to degenerates the female müllerian ducts

Testosterone converted to a potent androgen **DHT** (dihydro testosterone) by **5 α reductase** enzyme .
DHT induces masculinization and development of urethra ,prostate and external genitalia (penis and scrotum)

